

Block-encoding K matrices A_k of size $2^n \times 2^n$

$K = 8$ blocks; n sweep = [2, 3, 4, 5, 6, 7]; phase_bitsize $b = 32$.

Approach 1 (SVD-based):

$A_k = U_k \Sigma_k V_k$. Block-encoded as

$$\left[\sum_k |k\rangle\langle k| (x) U_k \right] \cdot \left[\text{diag}(\text{sigma}) + \text{ancilla rotation} \right] \cdot \left[\sum_k |k\rangle\langle k| (x) V_k \right].$$

- Interferometer cost for U_k and V_k :

`BlockUnitaryInterferometerSynthesisQROAM(n_blocks=K, n_rows=2^n, n_layers=2^n).`

- Diagonal cost: `QROAMClean(data_shape=(K, 2^n), b) + ctrl-AddIntoPhaseGrad(b,b) + QROAMCleanAdjoint.`

- Toffoli total = $2 * \text{interferometer} + \text{diagonal}$.

- Qubit total $\sim \max(\text{interferometer}, \text{diagonal}) + 1$ (block-encoding ancilla).

Approach 2 (direct dilation):

Each A_k is completed (Stinespring) to a unitary W_k of size $2^{(n+1)} \times 2^{(n+1)}$.

Synthesize $\sum_k |k\rangle\langle k| (x) W_k$ via one interferometer on $(n+1)$ system qubits:

`BlockUnitaryInterferometerSynthesisQROAM(n_blocks=K, n_rows=2^(n+1), n_layers=2^(n+1)).`

All resource counts come from Qualtran's `get_cost_value(QECGatesCost())`, via `utils.get_Toffoli_counts` and `utils.get_qubit_counts`. Each cost is optimized over `log_block_size` in `0..11`.

Two optimization modes per approach are shown: Toffoli-optimal and qubit-optimal.

Fits $y = c * 2^{(\alpha * n)}$:

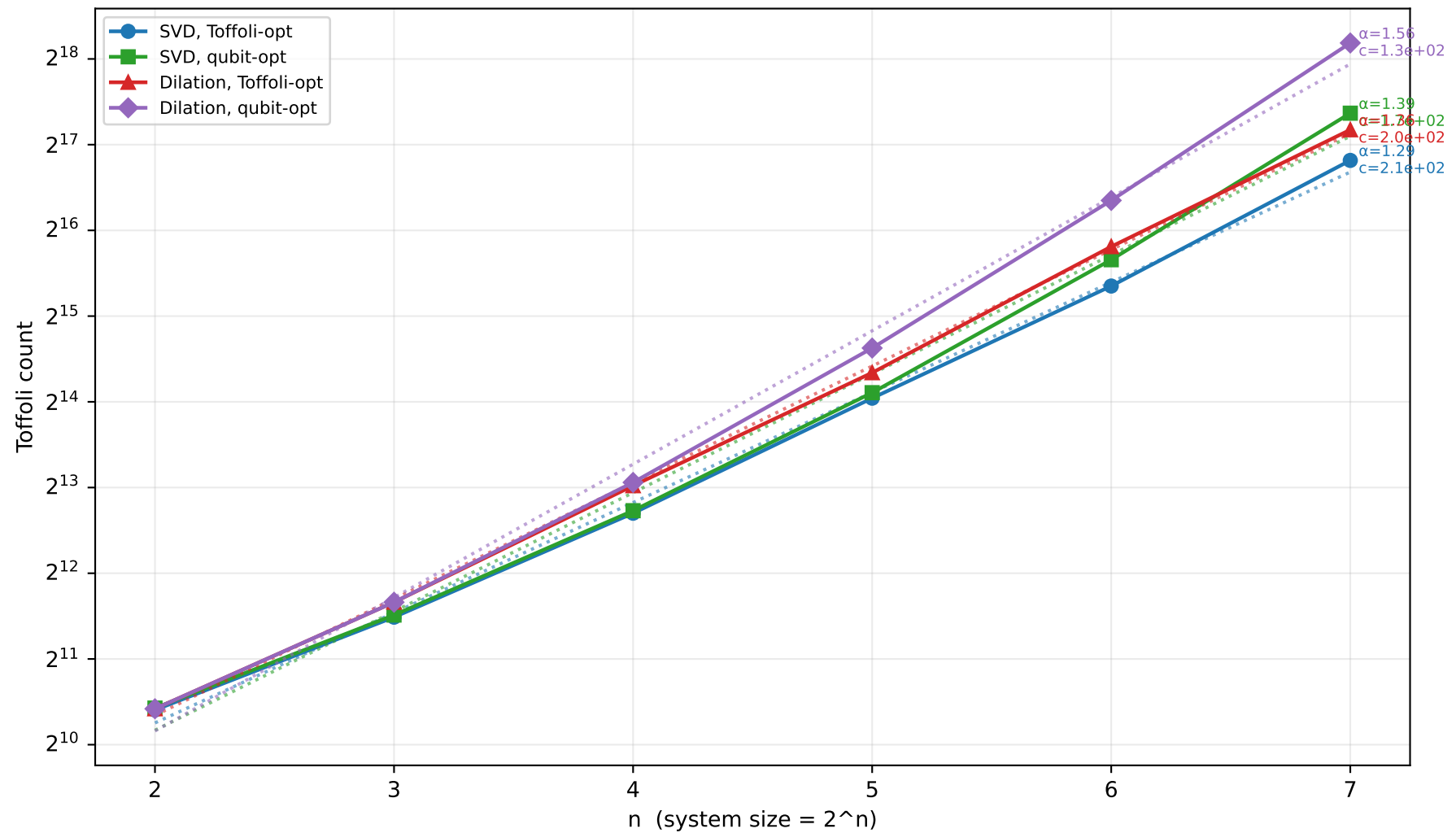
SVD, Toffoli-opt T: $\alpha=1.285$, $c=2.055e+02$; Q: $\alpha=0.536$, $c=6.958e+01$

SVD, qubit-opt T: $\alpha=1.386$, $c=1.684e+02$; Q: $\alpha=0.026$, $c=1.003e+02$

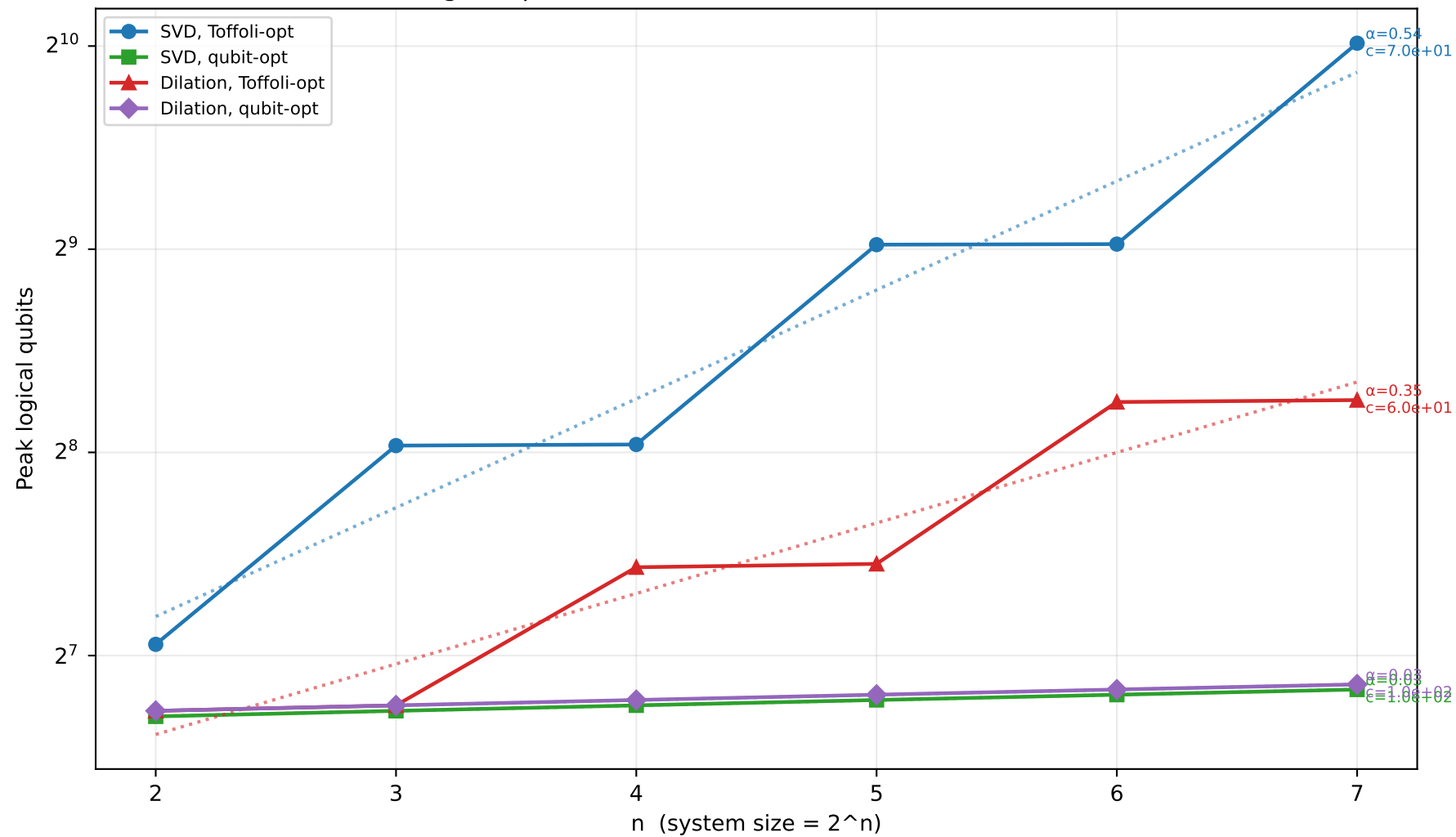
Dilation, Toffoli-opt T: $\alpha=1.358$, $c=1.975e+02$; Q: $\alpha=0.347$, $c=6.045e+01$

Dilation, qubit-opt T: $\alpha=1.556$, $c=1.323e+02$; Q: $\alpha=0.026$, $c=1.023e+02$

Toffoli count: SVD vs direct dilation (K=8 blocks, b=32)



Peak logical qubits: SVD vs direct dilation (K=8 blocks, b=32)



Resource counts (K=8, b=32)

n	dim	SVD T-opt T	SVD T-opt Q	SVD Q-opt T	SVD Q-opt Q	Dil T-opt T	Dil T-opt Q	Dil Q-opt T	Dil Q-opt Q
2	4	1,356	133	1,377	104	1,369	106	1,369	106
3	8	2,872	262	2,921	106	3,241	108	3,241	108
4	16	6,656	263	6,793	108	8,314	173	8,537	110
5	32	16,874	520	17,641	110	20,730	175	25,305	112
6	64	41,786	521	51,689	112	57,468	304	83,481	114
7	128	115,406	1034	169,065	114	147,836	306	298,265	116